

# MINOR PROJECT – 3

## Terraform AWS Infrastructure Setup

Infrastructure Provisioning, Networking and EC2 Deployment

|                        |                                    |
|------------------------|------------------------------------|
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| <b>Project</b>         | Terraform AWS Infrastructure Setup |
| <b>Platform</b>        | Amazon Web Services                |
| <b>Submission Date</b> | 07 September 2026                  |

This report presents the AWS infrastructure provisioned using Terraform, including the VPC, public subnet, networking components, security group and EC2 instance.

## 1. Introduction

Terraform is an Infrastructure as Code (IaC) tool used to define and provision cloud infrastructure through configuration files. In this project, Terraform was used with Amazon Web Services to create a VPC, public subnet, Internet Gateway, route table, security group and an EC2 instance. The deployed EC2 instance was then accessed successfully through SSH.

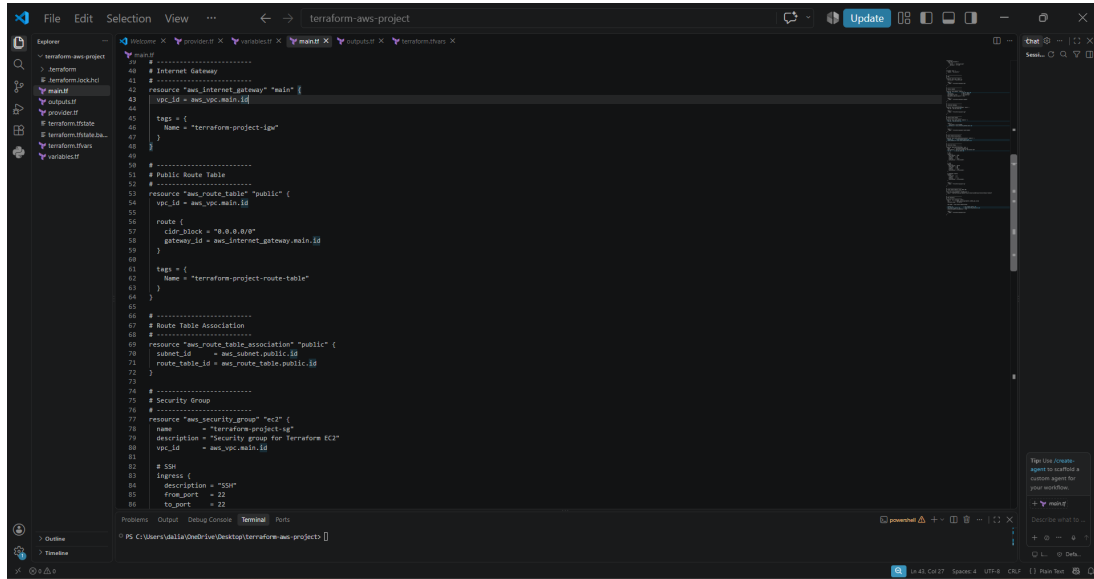
## 2. Objectives

- Create and configure an AWS VPC using Terraform.
- Create a public subnet and configure internet connectivity.
- Create an Internet Gateway and public route table.
- Configure a security group for SSH and HTTP access.
- Launch an EC2 instance using Terraform.
- Connect to the EC2 instance through SSH and verify access.

## 3. Project Configuration

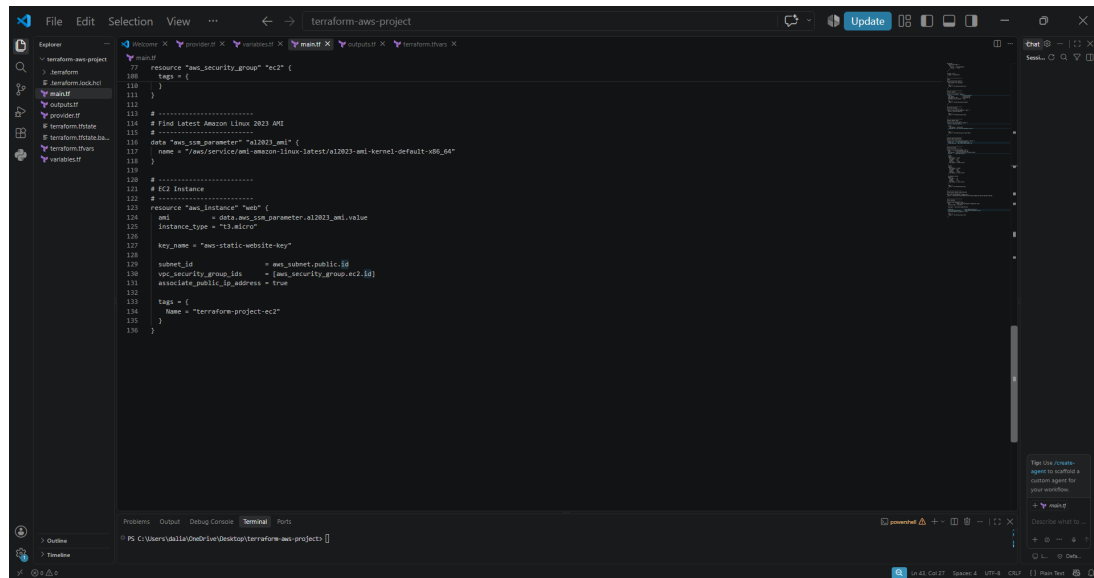
| Resource         | Configuration                                                 |
|------------------|---------------------------------------------------------------|
| VPC              | terraform-project-vpc; CIDR 10.0.0.0/16                       |
| Public Subnet    | terraform-project-subnet; CIDR 10.0.1.0/24; ap-south-1a       |
| Internet Gateway | terraform-project-igw                                         |
| Route Table      | terraform-project-route-table; 0.0.0.0/0 via Internet Gateway |
| Security Group   | terraform-project-sg; SSH 22 and HTTP 80                      |
| EC2 Instance     | terraform-project-ec2; t3.micro; Amazon Linux 2023            |

## 4. Terraform Implementation Evidence



```
40 # Internet Gateway
41 resource "aws_internet_gateway" "main" {
42   vpc_id = aws_vpc.main.id
43 }
44
45 # Public Route Table
46 resource "aws_route_table" "public" {
47   vpc_id = aws_vpc.main.id
48
49   route {
50     cidr_block = "0.0.0.0/0"
51     gateway_id = aws_internet_gateway.main.id
52   }
53
54   tags = {
55     Name = "terraform-project-route-table"
56   }
57 }
58
59 # Route Table Association
60 resource "aws_route_table_association" "public" {
61   subnet_id = aws_subnet.public.id
62   route_table_id = aws_route_table.public.id
63 }
64
65 # Security Group
66 resource "aws_security_group" "ec2" {
67   name = "terraform-project-sg"
68   description = "Security group for Terraform EC2"
69   vpc_id = aws_vpc.main.id
70
71   ingress {
72     description = "SSH"
73     from_port = 22
74     to_port = 22
75     protocol = "tcp"
76     cidr_blocks = ["0.0.0.0/0"]
77   }
78 }
```

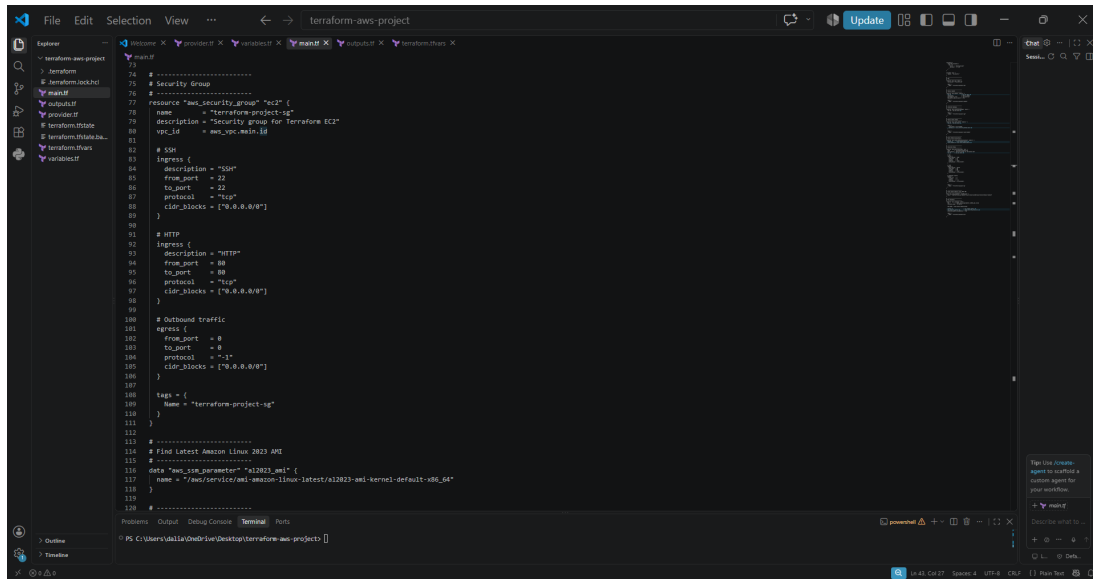
Terraform configuration – VPC, subnet and networking resources



```
77 resource "aws_security_group" "ec2" {
78   name = "terraform-project-sg"
79   description = "Security group for Terraform EC2"
80   vpc_id = aws_vpc.main.id
81
82   ingress {
83     description = "SSH"
84     from_port = 22
85     to_port = 22
86     protocol = "tcp"
87     cidr_blocks = ["0.0.0.0/0"]
88   }
89 }
90
91 # EC2 Instance
92 resource "aws_instance" "web" {
93   ami = data.aws_ssm_parameter.al2023_ami.value
94   instance_type = "t3.micro"
95   key_name = "aws-static-website-key"
96   subnet_id = aws_subnet.public.id
97   vpc_security_group_ids = [aws_security_group.ec2.id]
98   associate_public_ip_address = true
99
100   tags = {
101     Name = "terraform-project-ec2"
102   }
103 }
```

Terraform configuration – route table, security group and EC2 resources

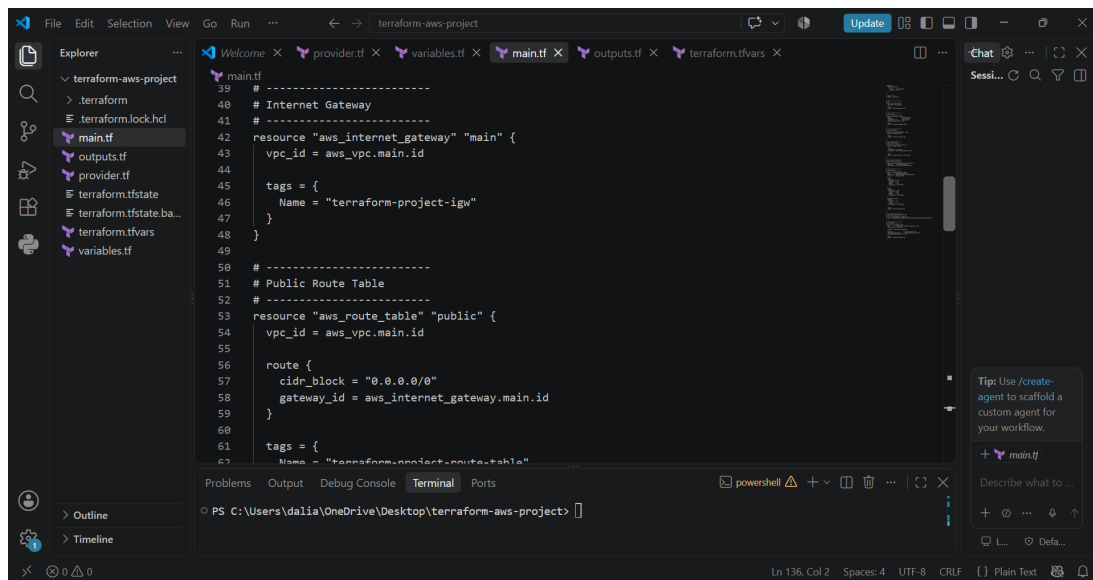
## 4. Terraform Implementation Evidence (continued)



This screenshot shows the VS Code editor with the 'main.tf' file open. The file contains Terraform configuration for an AWS Security Group, an Ingress rule for SSH, an HTTP rule, and an Outbound traffic rule. The configuration is as follows:

```
73 #
74 # Security Group
75 #
76 resource "aws_security_group" "ec2" {
77   name        = "terraform-project-sg"
78   description = "Security group for Terraform EC2"
79   vpc_id      = aws_vpc.main.id
80 }
81
82 # SSH
83 ingress {
84   description = "SSH"
85   from_port   = 22
86   to_port     = 22
87   protocol    = "tcp"
88   cidr_blocks = ["0.0.0.0/0"]
89 }
90
91 # HTTP
92 ingress {
93   description = "HTTP"
94   from_port   = 80
95   to_port     = 80
96   protocol    = "tcp"
97   cidr_blocks = ["0.0.0.0/0"]
98 }
99
100 # Outbound traffic
101 egress {
102   from_port   = 0
103   to_port     = 0
104   protocol    = "-1"
105   cidr_blocks = ["0.0.0.0/0"]
106 }
107
108 tags = {
109   Name = "terraform-project-sg"
110 }
111 }
112
113 # Find latest Amazon Linux 2023 AMI
114 data "aws_ami_parameter" "al2023_ami" {
115   name = "aws-service-ami-amazon-linux-latest/al2023-ami-kernel-default-hfs_64"
116 }
117
118 #
```

Terraform configuration – EC2 and networking configuration

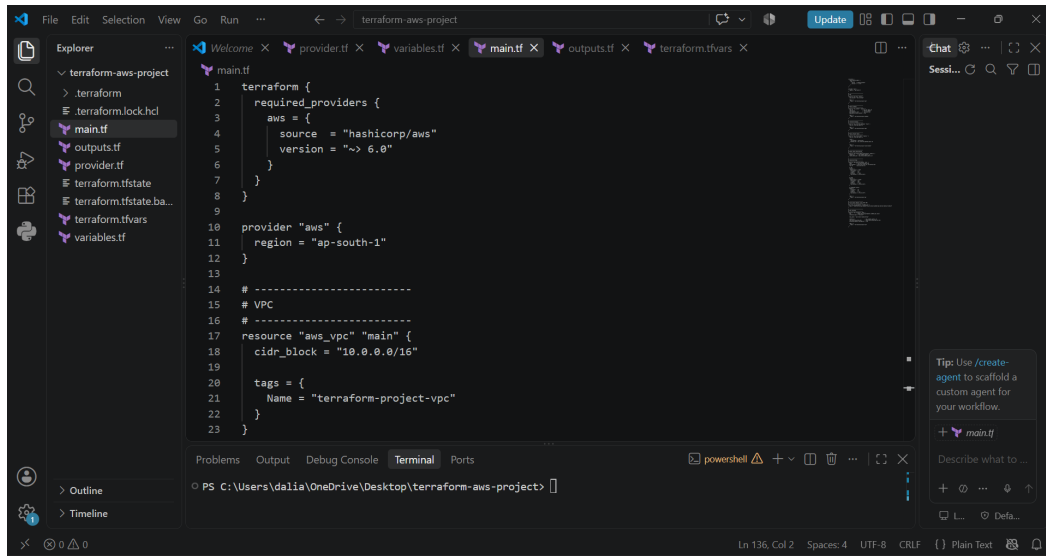


This screenshot shows the VS Code editor with the 'main.tf' file open. The file contains Terraform configuration for an Internet Gateway, a Public Route Table, and a route. The configuration is as follows:

```
39 #
40 # Internet Gateway
41 #
42 resource "aws_internet_gateway" "main" {
43   vpc_id = aws_vpc.main.id
44 }
45
46 tags = {
47   Name = "terraform-project-igw"
48 }
49
50 #
51 # Public Route Table
52 #
53 resource "aws_route_table" "public" {
54   vpc_id = aws_vpc.main.id
55 }
56
57 route {
58   cidr_block = "0.0.0.0/0"
59   gateway_id = aws_internet_gateway.main.id
60 }
61
62 tags = {
63   Name = "terraform-project-route-table"
64 }
```

Terraform project files and configuration in VS Code

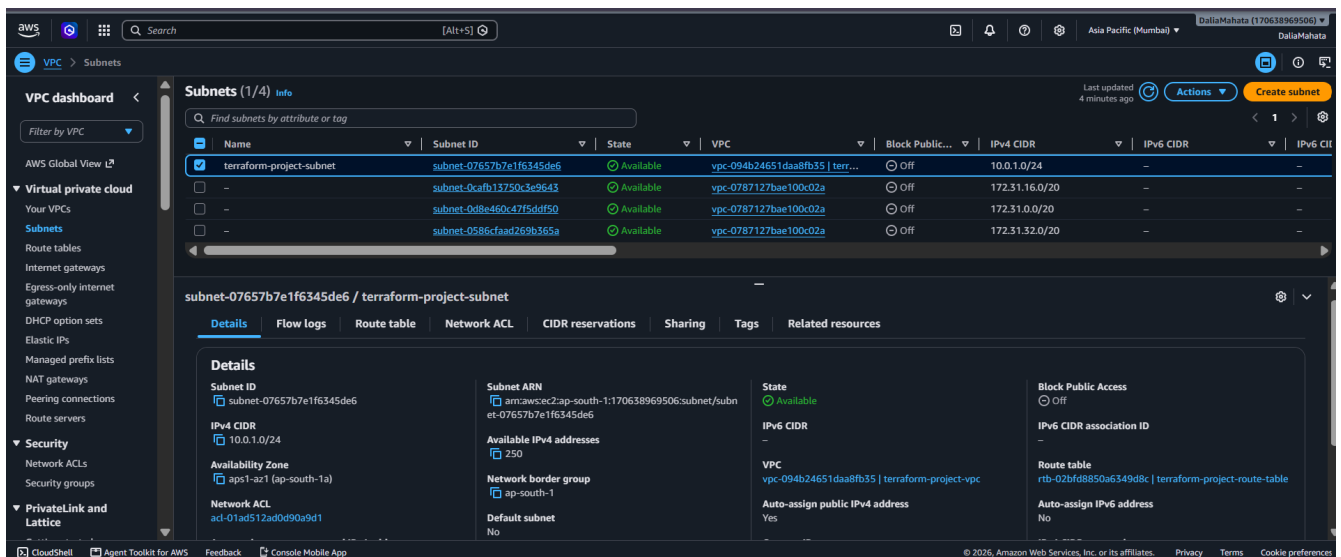
## 5. AWS Infrastructure Verification



The screenshot shows a code editor with a file explorer on the left displaying a project structure for 'terraform-aws-project'. The main editor window shows the 'main.tf' file with Terraform configuration for an AWS VPC. The configuration includes required providers, the AWS provider setup for the 'ap-south-1' region, and a resource definition for 'aws\_vpc' with a 'main' tag. The VPC CIDR block is set to '10.0.0.0/16'. A chat window is visible on the right side of the editor.

```
1 terraform {
2   required_providers {
3     aws = {
4       source = "hashicorp/aws"
5       version = "~> 6.0"
6     }
7   }
8 }
9
10 provider "aws" {
11   region = "ap-south-1"
12 }
13
14 # -----
15 # VPC
16 # -----
17 resource "aws_vpc" "main" {
18   cidr_block = "10.0.0.0/16"
19
20   tags = {
21     Name = "terraform-project-vpc"
22   }
23 }
```

AWS Console – VPC / Terraform project configuration



The screenshot shows the AWS Management Console interface for the 'Subnets' page. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, Subnets, Route tables, Internet gateways, Egress-only internet gateways, DHCP option sets, Elastic IPs, Managed prefix lists, NAT gateways, Peering connections, Route servers, Security, Network ACLs, Security groups, PrivateLink and Lattice. The main content area displays a table of subnets for the 'terraform-project-subnet' VPC. The table has columns for Name, Subnet ID, State, VPC, Block Public Access, IPv4 CIDR, IPv6 CIDR, and IPv6 CIDR. Below the table, the details for the selected subnet 'subnet-07657b7e1f6345de6' are shown, including Subnet ARN, Available IPv4 addresses, Network border group, Network ACL, and Block Public Access settings.

| Name                     | Subnet ID                | State     | VPC                             | Block Public... | IPv4 CIDR      | IPv6 CIDR | IPv6 CIDR |
|--------------------------|--------------------------|-----------|---------------------------------|-----------------|----------------|-----------|-----------|
| terraform-project-subnet | subnet-07657b7e1f6345de6 | Available | vpc-094b24651daa8fb35   terr... | Off             | 10.0.1.0/24    | -         | -         |
| -                        | subnet-0cafb13750c3e9643 | Available | vpc-0787127bae100c02a           | Off             | 172.31.16.0/20 | -         | -         |
| -                        | subnet-0d8e460c47f5dd50  | Available | vpc-0787127bae100c02a           | Off             | 172.31.0.0/20  | -         | -         |
| -                        | subnet-0586cfaad269b365a | Available | vpc-0787127bae100c02a           | Off             | 172.31.32.0/20 | -         | -         |

| subnet-07657b7e1f6345de6 / terraform-project-subnet |                                                       |                                 |                                                                     |
|-----------------------------------------------------|-------------------------------------------------------|---------------------------------|---------------------------------------------------------------------|
| Details                                             |                                                       |                                 |                                                                     |
| Subnet ID                                           | subnet-07657b7e1f6345de6                              | Subnet ARN                      | arn:aws:ec2:ap-south-1:170638969506:subnet/subnet-07657b7e1f6345de6 |
| IPv4 CIDR                                           | 10.0.1.0/24                                           | Available IPv4 addresses        | 250                                                                 |
| Availability Zone                                   | aps1-az1 (ap-south-1a)                                | Network border group            | ap-south-1                                                          |
| Network ACL                                         | acl-01ad512ad0d90a9d1                                 | Default subnet                  | No                                                                  |
| State                                               | Available                                             | IPv6 CIDR                       | -                                                                   |
| Block Public Access                                 | Off                                                   | VPC                             | vpc-094b24651daa8fb35   terraform-project-vpc                       |
| IPv6 CIDR association ID                            | -                                                     | Auto-assign public IPv4 address | Yes                                                                 |
| Route table                                         | rtb-02bfd8850a6349d8c   terraform-project-route-table | Auto-assign IPv6 address        | No                                                                  |

AWS Console – Public subnet: terraform-project-subnet

## 5. AWS Infrastructure Verification (continued)

The screenshot shows the AWS Management Console for the 'Security Groups' page. The left sidebar contains navigation options like 'VPC dashboard', 'Virtual private cloud', 'Subnets', 'Route tables', 'Internet gateways', 'Egress-only internet gateways', 'DHCP option sets', 'Elastic IPs', 'Managed prefix lists', 'NAT gateways', 'Peering connections', 'Route servers', 'Security', 'Network ACLs', 'Security groups', and 'PrivateLink and Lattice'. The main content area displays a table of security groups. The 'terraform-project-sg' group is selected, and its details are shown below the table.

| Name                 | Security group ID    | Security group name  | VPC ID                | Description                             | Owner        |
|----------------------|----------------------|----------------------|-----------------------|-----------------------------------------|--------------|
| terraform-project-sg | sg-05af8a31597dc35bf | terraform-project-sg | vpc-094b24651daa8fb35 | Security group for Terraform EC2        | 170638969506 |
| -                    | sg-0c18c54fa187c17a  | launch-wizard-2      | vpc-0787127bae100c02a | launch-wizard-2 created 2026-08-22T1... | 170638969506 |
| -                    | sg-0ab78cd6e728a3553 | default              | vpc-094b24651daa8fb35 | default VPC security group              | 170638969506 |
| -                    | sg-03aa4504beb7cb6f4 | launch-wizard-1      | vpc-0787127bae100c02a | launch-wizard-1 created 2026-08-16T1... | 170638969506 |
| -                    | sg-08f8ede9d4ca6a9b9 | default              | vpc-0787127bae100c02a | default VPC security group              | 170638969506 |

Below the table, the details for 'sg-05af8a31597dc35bf - terraform-project-sg' are shown. The details include the security group name, ID, description, VPC ID, owner, inbound rules count (2), and outbound rules count (1).

AWS Console – Security group terraform-project-sg

The screenshot shows the AWS Management Console for the 'Instances' page. The left sidebar contains navigation options like 'EC2', 'Dashboard', 'AWS Global View', 'Events', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', 'Images', 'AMIs', 'AMI Catalog', 'Elastic Block Store', 'Volumes', 'Snapshots', and 'Lifecycle Manager'. The main content area displays a table of instances. The 'terraform-project-ec2' instance is selected, and its details are shown below the table.

| Name            | Instance ID         | Instance state | Instance type | Status check      | Availability Zone | Public IPv4 DNS          | Public IPv4 ... | Elastic IP | IPv6 IPs |
|-----------------|---------------------|----------------|---------------|-------------------|-------------------|--------------------------|-----------------|------------|----------|
| AWS-Static-W... | i-078294a4d8f6421df | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-13-203-158-192.ap... | 13.203.158.192  | -          | -        |
| ject-ec2        | i-01dbdf63d7889ba82 | Running        | t3.micro      | 3/3 checks passed | ap-south-1a       | -                        | 13.206.195.184  | -          | -        |

Below the table, the details for 'i-01dbdf63d7889ba82 (terraform-project-ec2)' are shown. The details include the instance ID, public IPv4 address, private IPv4 addresses, instance state, private IP DNS name, instance type, and elastic IP addresses.

AWS Console – EC2 instance terraform-project-ec2

## 6. Terraform Validation and Deployment Results

The Terraform configuration was initialized, validated, planned and applied successfully. The final deployment created seven AWS resources without changes or destruction.

Terraform commands used:

```
terraform init
terraform validate
terraform plan
terraform apply
terraform output
```

Terraform output:

```
ec2_instance_id = "i-01dbdf63d7889ba82"
ec2_public_ip   = "13.206.195.184"
security_group_id = "sg-05af8a31597dc35bf"
subnet_id       = "subnet-07657b7e1f6345de6"
vpc_id          = "vpc-094b24651daa8fb35"
```

## 7. Security Group and EC2 Verification

The screenshot shows the AWS Management Console interface for Security Groups. The left sidebar contains navigation links for VPC, Virtual private cloud, and Security. The main content area displays a list of security groups. The 'terraform-project-sg' is selected, and its details are shown, including inbound rules. The inbound rules table shows two rules: one for HTTP port 80 and one for SSH port 22.

| Name                 | Security group ID    | Security group name  | VPC ID                | Description                             | Owner        |
|----------------------|----------------------|----------------------|-----------------------|-----------------------------------------|--------------|
| terraform-project-sg | sg-05af8a31597dc35bf | terraform-project-sg | vpc-094b24651daa8fb35 | Security group for Terraform EC2        | 170638969506 |
| -                    | sg-0cf8c54fa18f7c17a | launch-wizard-2      | vpc-0787127bae100c02a | launch-wizard-2 created 2026-08-22T1... | 170638969506 |
| -                    | sg-0ab78cd6e728a3553 | default              | vpc-094b24651daa8fb35 | default VPC security group              | 170638969506 |
| -                    | sg-03aa4504beb7cb6f4 | launch-wizard-1      | vpc-0787127bae100c02a | launch-wizard-1 created 2026-08-16T1... | 170638969506 |
| -                    | sg-08f8edeb94ca6a9b9 | default              | vpc-0787127bae100c02a | default VPC security group              | 170638969506 |

| Name | Security group rule ID | IP version | Type | Protocol | Port range | Source    | Description |
|------|------------------------|------------|------|----------|------------|-----------|-------------|
| -    | sgr-02499193450fc135   | IPv4       | HTTP | TCP      | 80         | 0.0.0.0/0 | HTTP        |
| -    | sgr-0f1188c717747b7ca  | IPv4       | SSH  | TCP      | 22         | 0.0.0.0/0 | SSH         |

Security group inbound rules – HTTP port 80 and SSH port 22 are configured.

## 8. SSH Connection Verification

The EC2 instance was successfully accessed through SSH using the configured key pair. The connection was verified with `whoami`, `hostname` and `pwd`. The session confirmed the user as `ec2-user` and the working directory as `/home/ec2-user`.

## 9. Conclusion

The project successfully demonstrates practical Infrastructure as Code using Terraform and AWS. The VPC and public subnet were created with the required networking components, a security group was configured for SSH and HTTP access, and an EC2 instance was launched in the Mumbai region. The instance reached a running state with successful status checks and was accessed through SSH. The collected Terraform and AWS Console evidence demonstrates that the infrastructure was provisioned and verified successfully.

## 10. References

- Amazon Web Services — Amazon VPC documentation
- Amazon Web Services — Amazon EC2 documentation
- HashiCorp Terraform documentation
- AWS Management Console
- Terraform AWS Provider documentation

— End of Report —